

feedback to refine the UI. Usability study for some critical products can go through three to four iterations before freezing on the final one. Over a period of time, this practice has matured and, with experience, it has generated two streams of output for designers:

1. Standard design inputs, which are common to most products, are related to UI components like keyboards, displays, knobs and levers, their layout and grouping, and can be categorised as must-have features.

2. Product specific inputs, which need special design inputs depending on product functionality (for example, touch based slider for volume control), need to be specially assessed by ID engineers.

In this article, we will see some of the must-have features specially related to keys and knobs, among others. These inputs will range from selection of parts to location and colour to be used. These inputs will help designers of small products and designers from small- and medium-size company designers to have a low-cost but acceptable UID. However, they need to be careful about one point; if their designs are applicable to safety-critical industrial systems and medical devices, getting help from professional designers is a must.

Basic usability engineering (UID)

When a human being interacts with a machine (product), there are three types of interactions that take place:

1. Human-machine
2. Human-workspace
3. Human-environment

A good UI will balance the impact of all these equally and ensure that the user is not adversely impacted while using the product.

Let us see what each of these interactions mean to designers.

Human-machine. This interaction involves the influence of the product on the user and his decisions on displays, controls, panel layouts, rate of information dissemination, etc. Essentially, it means how users react

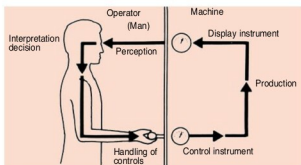


Fig. 1: Relationship between a human being and a machine

to the product's UI when they are interacting with the product.

Human-workspace. This interaction involves the user, his or her posture, position, how much he or she has to reach out to use the product, product size, structure of the product, and so on. Essentially, this means the impact on the user, especially physically, while using the product.

Human-environment. This interaction involves the behaviour of the user based on the working environment like light, temperature, sound (acoustics), noise, ventilation and radiation. Essentially, this deals with the environment in which the product works and how that will impact the user.

A good example can be an industrial control system with an audio alarm on a noisy shop floor. Normally, the user in this environment will use earplugs to muffle the sound (noise). So if the product has to work in an environment like that, he or she needs to use a visual alarm along with an audio alarm.

UI-human interaction chain

A good UID is a closed-loop feedback design. Fig. 1 shows the relationship between a human and a machine.

When this loop gets disrupted, external errors happen. While we normally call this human error, the trigger is from the external world. This means that the display and controls of a system (product) need to be designed in a way that induced errors are minimised or removed completely, wherever possible.

With this background, let us look

at the types of user interactions and types of users for better understanding of UI design.

Human-machine interaction types

1. Conventional systems or products with switches, keyboards, displays and alarms

2. Computer interface, which is also

called human-computer interaction (HCI); typically uses keyboards, touchscreens and monitor displays

Types of users. Users are classified into three categories:

Novice. Someone who is a first-time user of the product (has less exposure to the product)

Expert. Someone who has used the product earlier (knows how to use the product)

Casual. Someone who can use the product; may not be as well-versed as an expert, but would have been exposed to other similar products

This means UID actually needs to cater to the types of usage, as well as the types of users, to be successful. Having understood the usage and user types, let us now see the actual design of a UI and components used for the same.

Essential UIs

For a design engineer, an important element of the UI are keys. Let us understand the design using these.

Controls (input devices). These include switches, potentiometers and valves, among others, and can be further classified into two groups:

Discrete controls. Each position in a switch represents a separate function, and switches are typically discrete controls. Fig. 2 shows some of the discrete control elements.

Continuous controls. Continuous controls change in value from a minimum to a maximum. Rotary valves, potentiometers and rotary switches are good examples of continuous controls. Fig. 3 shows some continuous control elements.